NÜWA-LIP: LANGUAGE-GUIDED IMAGE INPAINTING WITH DEFECT-FREE VQGAN

Anonymous authors
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ABSTRACT

Language-guided image inpainting aims to fill in the defective regions of an image under the guidance of text while keeping non-defective regions unchanged. However, the encoding process of existing models suffers from either receptive spreading of defective regions or information loss of non-defective regions, giving rise to visually unappealing inpainting results. To address the above issues, this paper proposes NÜWA-LIP by incorporating defect-free VQGAN (DF-VQGAN) with multi-perspective sequence to sequence (MP-S2S). In particular, DF-VQGAN introduces relative estimation to control receptive spreading and adopts symmetrical connections to protect information. MP-S2S further enhances visual information from complementary perspectives, including both low-level pixel and high-level token. Experiments show that DF-VQGAN performs much more robustly than VQGAN. To evaluate language-guided inpainting, we built up three open-domain benchmarks, where NÜWA-LIP is also superior to recent strong baselines.

1 INTRODUCTION

Image inpainting, which aims to fill missing pixels in the defective regions of a corrupted image to make the completed image photo-realistic, is as ancient as art itself (Bertalmio et al., 2000). It is a representative task in computer vision (Peng et al., 2021; Zhao et al., 2020; Zheng et al., 2019) with real-world applications, such as image manipulation, image completion, object removal, etc.

With the rapid progress in vision-language learning, language-guided image inpainting has become an encouraging topic. Such systems perform inpainting using not only given pixels but also natural language instructions, and can be implemented based on GAN models (Bau et al., 2021) or diffusion models (Esser et al., 2021a). Recently, autoregressive model-based multimodal pre-training has exhibited strong visual synthesis capabilities, such as Stable Diffusion (Rombach et al., 2022), Parti (Yu et al., 2022) and NÜWA (Wu et al., 2021a). In particular, NÜWA has shown a good zero-shot text-guided image manipulation capability, opening the possibility of such a generative pre-training mechanism for language-guided image inpainting.

However, existing pre-trained models encode the image in the same way for both known non-defective regions and defective regions. This may lead to fusing uncertain information from defective regions to the representations of non-defective regions and limits the accuracy of modeling non-defective regions. We name this Receptive Spreading of the defective region. Moreover, in the VQVAE-based model, the known non-defective regions are hard to be reconstructed exactly the same as the original image due to the compression of the image to discrete tokens. We name this Information Loss of the non-defective region (see Fig. 1).

To solve these problems, we propose NÜWA-LIP by leveraging a novel defect-free VQGAN (DF-VQGAN) and multi-perspective sequence to sequence (MP-S2S). In contrast to VQGAN (Esser et al., 2021b), DF-VQGAN incorporates the relative estimation for decoupling non-defective regions from defective regions to control receptive spreading. To protect the information of non-defective regions, symmetrical connections replenish the lost information of non-defective information from the encoding procedure. Additionally, MP-S2S further enhances visual information from complementary perspectives, including both low-level pixel and high-level token under the text guidance.

Furthermore, we build three evaluation datasets for language-guided image inpainting and perform a comprehensive comparison with existing works. Experiments show that NÜWA-LIP can achieve

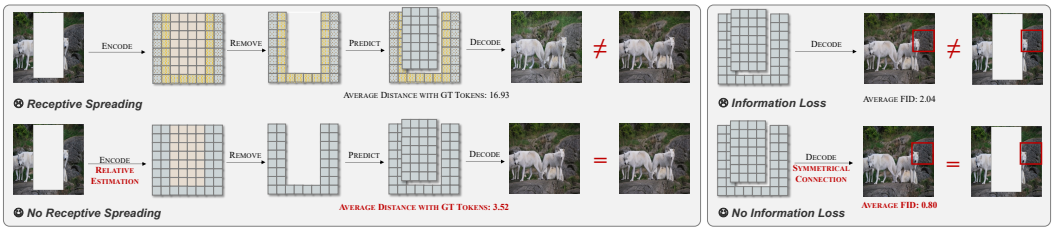


Figure 1: **Receptive Spreading and Information Loss.** For Receptive spreading, we can observe a hue change and a blurry boundary of known non-defective regions in the image generated. For Information loss, we can observe some changes in non-defective regions in the image generated.

state-of-the-art results on these three benchmarks. We also conduct ablation studies to assess the effect of different components in NÜWA-LIP.

To sum up, the major contributions of this paper involve:

- We propose a DF-VQGAN, which introduces relative estimation to control receptive spreading and adopts symmetrical connections to protect the information of non-defective regions.
- We propose a multi-perspective sequence to sequence, which enhances visual information from complementary perspectives, including both low-level pixel and high-level token representation under the text guidance.
- We build three evaluation datasets for language-guided image inpainting. Experiments show that NÜWA-LIP achieved state-of-the-art results in comparison to several strong baselines.

2 Related Work

Vector Quantized Variational AutoEncoder Vector Quantized Variational AutoEncoder is a VAE model which can compress the continuous information to discrete latent variables (Oord et al., 2017). Several works, such as VQVAE-2 (Razavi et al., 2019), try to decode the image at a more fine-grained level. To decode a more vivid image, VQGAN (Esser et al., 2021b) uses a GAN model to make the decoded images indistinguishable from the real images. However, existing models generally focus on normal images. As for the images containing defective regions, especially in inpainting scenarios, the defective regions in the image will affect all discrete latent variables due to receptive spreading. This may lead to color cast or faults in the decoded image. Meanwhile, the information loss will cause the modification of the non-defective region, decreasing the quality of inpainting.

Language-guided Image Inpainting Language-guided image inpainting, as a subfield of text-image synthesis, has attracted more and more attention. This task aims to fill in the defective regions of the image with text guidance that describes the content of the full image. Analogous to unconditional image inpainting (Suvorov et al., 2022; Zhao et al., 2021; Yu et al., 2019), some works, such as (Bau et al., 2021; Wu et al., 2021b; Zhang et al., 2020) introduced generative adversarial networks to handle generic real images. Recently, many works introduced pre-training on large-scale data to improve inpainting ability in a general domain instead of a specific domain. Several projects, such as ImageBART (Esser et al., 2021a) and GLIDE (Nichol et al., 2021), perform this task with pre-training diffusion model. Following DALL-E (Ramesh et al., 2021), some other works like NÜWA (Wu et al., 2021a) utilize the autoregressive model. However, existing models usually suffer from the problems of receptive spreading of defective regions or information loss of non-defective regions. Moreover, modeling the image in a single way limits the quality of the inpainting results.

3 Method

3.1 Problem Formulation

Given an input image $x \in \mathbb{R}^{W \times H \times C}$ with width W, height H and channel C, a mask matrix $m \in \{0,1\}^{W \times H}$ with value 1 denoting the defective regions, and a piece of natural text t, the language-guided image inpainting aims to repair the defective regions under the guidance of the

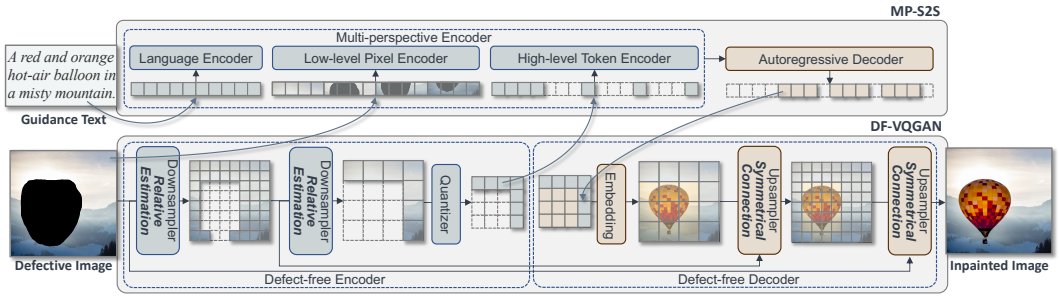


Figure 2: **Overview of NÜWA-LIP during inference.** The encoder of DF-VQGAN will generate the tokens of the image. Then, tokens corresponding to defective regions will be replaced with tokens predicted by the MP-S2S. The decoder of DF-VQGAN will generate the image based on these tokens.

text and output a new image $y \in \mathbb{R}^{W \times H \times C}$. Hereafter, we call the input x **defective image** and the expected output y **ground-truth image**.

In the Bayesian framework, the language-guided image inpainting task can be defined as maximizing the log posterior probability[1], as denoted in Eq. (1):

$$\log p(y|x, m, t; \omega, \phi, \psi) = \log \frac{p(y|z, x, m; \psi)p(z|x, m, t; \phi)}{p(z|y, x, m; \omega)}, \tag{1}$$

where ω, ϕ, ψ denote model parameters. z denotes the latent tokens. By taking the expectation w.r.t a auxiliary density $z \sim q(z|y, x, m; \phi)$ on both sides, the right side of Eq. (1) can be rewritten as:

$$\begin{aligned} \mathbb{E}_{z\sim q(z|y,x,m;\phi)}[\log p(y|z, x, m; \psi)] - \mathbb{KL}_{z\sim q(z|y,x,m;\phi)}\left[q(z|y, x, m; \phi)||p(z|x, m, t; \phi)\right] \\ + \mathbb{KL}_{z\sim q(z|y,x,m;\phi)}\left[q(z|y, x, m; \phi)||p(z|y, x, m; \omega)\right]. \end{aligned} \tag{2}$$

According to VAE, since the third Kullback-Leibler divergence term is always greater than 0, we only need to maximize the first two terms denoted as the Evidence Lower BOund (ELBO). The first expectation term is the reconstruction loss of the inpainted image. The second term is a Kullback-Leibler divergence loss, which constrains the conditional distribution of the latent tokens generated by the VAE encoder should be close to that generated by the auxiliary probability density. As Fig. 2, we will introduce how we model the first term with a defect-free VQGAN (DF-VQGAN) in Sec. 3.2 and the second term with a multi-perspective sequence to sequence (MP-S2S) in Sec. 3.3. Note the term of mask m and defective image x in ELBO. This explains the motivation and feasibility to introduce such two terms into DF-VQGAN.

3.2 DF-VQGAN

The modeling of $\mathbb{E}_{z\sim q(z|y,x,m;\phi)}[\log p(y|z, x, m; \psi)]$ in Eq. (2) can be split into a VAE encoder of $q(z|y, x, m; \phi)$ and a VAE decoder of $p(y|z, x, m; \psi)$. The probabilistic density function $q(z|y, x, m; \phi)$ shows that the probability distribution of z should be conditioned given three variables: inpainted image y, defective image x and mask matrix m. In other words, the latent tokens z should not only represent the inpainted image but also be sensitive to defective images with masked regions. To achieve this, we propose DF-VQGAN, a defect-free VAE model.

Defect-free Encoder To obtain an equivalent latent token sequence $z^y \in \mathbb{R}^{w \times h \times n_z}$ of the ground-truth inpainted image y, we feed y to encoder E, which typically consists of several normal operations, such as attentions or convolutions:

$$z^y = E(y), \tag{3}$$

where $w \times h$ is the length of the sequence and n_z is the dimensionality of each latent token.

Referring to white defects of the VQGAN column in Fig. 4, the defective regions in x will affect non-defective regions during the encoding, which makes it different from encoding y. Mathematically, if we feed an image x into a convolutional network, the number of affected latent tokens is the number

[1]We assume t is conditionally independent of y given z, x, m.

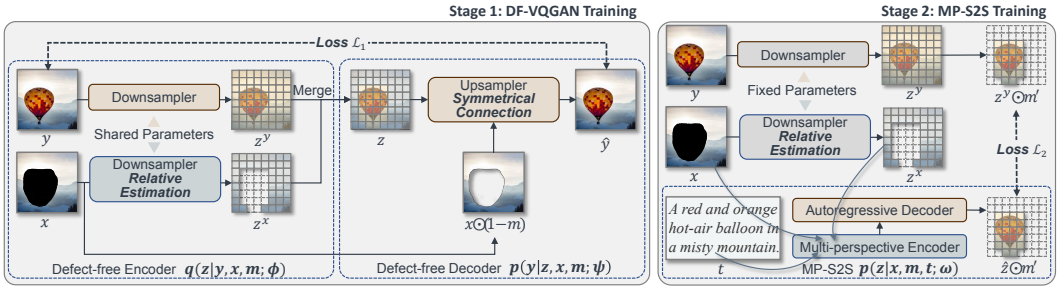


Figure 3: **Overview of NÜWA-LIP during two-stage training.** We optimize the whole NÜWA-LIP with two-stage training. Since we only have ground-truth inpainted image y in training data, we generate corresponding defective image x with random mask m. For Stage 1, we optimize DF-VQGAN via Eq. (12) with e and e^{DF} shared parameters. We use ground-truth inpainted image y and defective image x as the input image. For Stage 2, we optimize MP-S2S via Eq. (13) with fixed DF-VQGAN. We use DF-VQGAN to generate the training target $z^y \odot m'$ and input latent tokens z^x.

of 1 in $\mathrm{MaxPool}(m)$, which uses the same stride and kernel size with the convolution. For the network with the attention or normalization operation, more latent tokens will be affected.

Could we only encode the non-defective regions in x to prevent the effects from defective regions? Thanks to the mask matrix, we could easily mask the defective parts for attention and convolution operations. Using the re-estimated mean and variance, the normalization[2] can be given by,

$$\mathrm{Norm}^{\mathrm{DF}}(x,m) = \frac{x - \frac{N}{N_m}\mathrm{E}[x]}{\sqrt{\frac{N-1}{N_m-1}\mathrm{Var}[x'] + \epsilon}}, \tag{4}$$

where N denotes the number of pixels in x, N_m denotes the number of defective pixels. x' denotes the x with the defective region fulfilled with $\frac{N}{N_m}\mathrm{E}[x]$. $\mathrm{E}[\cdot]$ and $\mathrm{Var}[\cdot]$ denotes expectation and variance. We named these defect-free operations with ***relative estimation***.

Then, we could replace the operations[3] in encoder E with ***relative estimation*** to get the defect-free encoder E^{DF} and carefully encode the defective image x:

$$z^x = E^{\mathrm{DF}}(x,m), \tag{5}$$

where $z^x \in \mathbb{R}^{w\times h\times n_z}$ is latent tokens of x without polluting the non-defective part.

To ensure the consistency of z^x and z^y, they are not trained separately. As Fig. 3, we merge them into a joint latent tokens sequence $z \in \mathbb{R}^{w\times h\times n_z}$ and optimize them jointly. By a simple $\mathrm{MaxPool}$, the mask matrix m can be resized to m' to fit the size of z^x and z^y. Then, we can use $m' \in \mathbb{R}^{w\times h}$ to filter the non-defective parts of z^x and to replace the corresponding parts in z^y:

$$\begin{aligned} m' &= \mathrm{MaxPool}(m), \\ z &= z^y \odot m' + z^x \odot (1 - m'). \end{aligned} \tag{6}$$

Note that the encoding is an iterative process and the input will be transformed into the final latent tokens gradually. During the process, the defective region m also evolves. Thus, we also extend Eq. (3, 5, 6) to whole iterative process. Empirically, encoder E and defect-free encoder E^{DF} can be divided into a series of downsamplers $\{e_i\}_{i=1}^T$ and $\{e_i^{\mathrm{DF}}\}_{i=1}^T$. Note that e_i^{DF} and e share the same parameters and their main difference is whether to use ***relative estimation***.

Now, we can suppress the adverse effect of defective regions whatever their shape or scale. Formally, we can obtain the merged final latent tokens $z = h_T$ with:

$$\begin{aligned} h_i^y &= e_i(h_{i-1}), \\ h_i^x &= e_i^{\mathrm{DF}}(h_{i-1}, m_{i-1}), \\ m_i &= \mathrm{MaxPool}_i(m_{i-1}), \\ h_i &= h_i^y \odot m_i + h_i^x \odot (1 - m_i), \end{aligned} \tag{7}$$

[2]To simplify the formulation, we use Layer Norm to reveal our method, but this equation can be extended easily to other normalization functions, such as Group Norm.

[3]For more descriptions of the replacement, see Appendix A.1.

where the stride and kernel size of $\mathrm{MaxPool}_i$ are the same as the stride of the convolution in e_i. Let $m_0 = m$ and $h_0 = y$ as the initial value of the process for training.

With a learnable codebook B, the discrete tokens $\tilde{z} \in \mathbb{R}^{w \times h}$ for DF-VQGAN is:

$$\tilde{z}_i = \arg\min_j ||z_i - B_j||^2. \tag{8}$$

By applying Eq. (8) to z^x or z^y, we can obtain discrete tokens $\tilde{z}^x$ or $\tilde{z}^y$ used for MP-S2S training.

Defect-free Decoder Following the VQGAN architecture, we first embed the discrete tokens $\tilde{z}$ by looking up the codebook via $z = B[\tilde{z}]$, which are then fed into the decoder G. Different from the encoding process in Eq. (5), we do not need to do defect-free operations, since there are no defective regions in z because this region has been implemented by either encoding from encoder E in training or prediction from the MP-S2S in inference.

Referring to Fig. 3, to avoid the information loss of non-defective region, we propose ***symmetrical connection***. Let $\hat{y} \in \mathbb{R}^{W \times H \times C}$ be the output of the decoder and τ be the mixture coefficient. The hidden state of the non-defective region in the encoder will be mixed with the output of the decoder:

$$\begin{aligned} \hat{y}' &= G(z), \\ \hat{y} &= (1 - m) \odot \frac{1}{\tau + 1}(\hat{y}' + \tau x) + m \odot \hat{y}'. \end{aligned} \tag{9}$$

Correspondingly, decoding is a reversed iterative process of encoding. We can divide decoder G into a series of upsamplers $d_1, d_2, ..., d_T$ that are symmetrical to downsamplers in E or E^{DF}. Similar with. Eq. (7), we extend Eq. (9) to whole iterative process and obtain the final output image $\hat{y} = \hat{h}_0$, which is reconstructed based on latent tokens z by,

$$\begin{aligned} \hat{h}'_i &= g_{i+1}(\hat{h}_{i+1}), \\ \hat{h}_i &= (1 - m_i) \odot \frac{1}{\tau_i + 1}(\hat{h}'_i + \tau_i h_i) + m_i \odot \hat{h}'_i, \end{aligned} \tag{10}$$

where we set $\tau_0 = \infty$ to remain non-defective region unchanged and $\tau_i = 1$ for $i \neq 0$ to obtain a smoother mixture. Symmetrically, the initial state of the process is $\hat{h}_T = z$.

Let $\mathrm{sg}[\cdot]$ be stop gradient function. Following VQGAN, the training objective is:

$$\begin{aligned} \mathcal{L}^V &= ||y - \hat{y}||_2^2 + ||\mathrm{sg}[z] - B[\tilde{z}]||_2^2 + ||\mathrm{sg}[B[\tilde{z}]] - z||_2^2, \\ \mathcal{L}^P &= ||Q(y) - Q(\hat{y})||_2^2, \\ \mathcal{L}^G &= \log D(y) + \log(1 - D(\hat{y})), \end{aligned} \tag{11}$$

where Q and D respectively denote CNN-based modules to obtain the conceptual representation of the image and discriminator. The overall learning objective of DF-VQGAN is:

$$\mathcal{L}_1 = \mathcal{L}^V + \mathcal{L}^P + \mathcal{L}^G. \tag{12}$$

3.3 MP-S2S

As noted in Sec. 3.2, we model the probabilistic density function $q(z|y, x, m; \phi)$ using the Kullback-Leibler divergence term $\mathbb{KL}_{z \sim q(z|y,x,m;\phi)}\left[q(z|y, x, m; \phi)||p(z|x, m, t; \phi)\right]$ by a DF-VQGAN encoder, while the key to solve this term relies on the modeling of density function $p(z|x, m, t; \phi)$. In this section, we propose a multi-perspective sequence to sequence (MP-S2S). MP-S2S encodes the information from three perspectives including the input text t, pixel-level defective image x, and its token-level representation z^x, and decodes the latent tokens of defective regions $\hat{z}$.

Multimodal Encoding For guidance text from language modalities, following BERT (Devlin et al., 2018), we tokenize the text with BPE and embed them to the representation sequence t, where $t_i \in \mathbb{R}^{n_t}$ denotes the representation of each token and n_t denotes the dimension of the representation. We encode token representation sequence t to get its representation by $c^{\mathrm{t}} = E^{\mathrm{t}}(t)$.

For the defective image, the model treats it from two perspectives, high-level token, and low-level pixel representation. For low-level pixel, following ViT (Dosovitskiy et al., 2020), we transform the

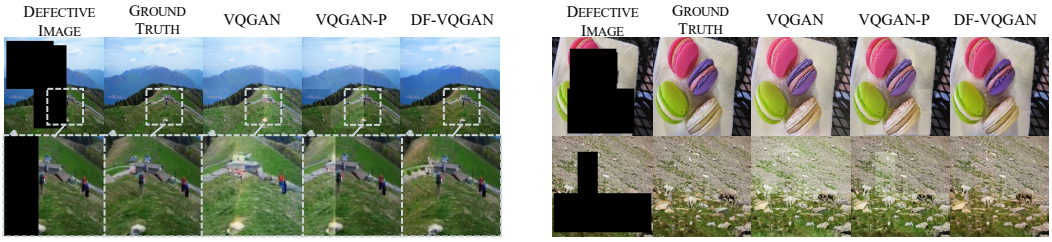


Figure 4: **Illustration on oracle inpainting.** Compared with VQGAN-P, we make a better transition between defective and non-defective regions and keep non-defective regions unchanged.

image to a sequence of non-overlapping patches $x^{\mathrm{p}} = (x_1^{\mathrm{p}}, x_2^{\mathrm{p}}, ...)$. Then, we directly encode the image patch sequence to get representation from low-level pixel perspective by $c^{\mathrm{l}} = E^{\mathrm{l}}(x^{\mathrm{p}})$. For high-level representation, we encode latent tokens z^x by $c^{\mathrm{h}} = E^{\mathrm{h}}(z^x)$. Note that tokens corresponding to defective regions, which can be located by m', will be replaced with a special trainable vector. We use Transformer encoder as the architecture of E^{h}, E^{l} and, E^{t}. Thus, the integrated representation at the hidden size dimension from two modalities can be formulated as $c = [c^{\mathrm{t}}; c^{\mathrm{l}}; c^{\mathrm{h}}]$.

Autoregressive Decoding The integrated representation c from two modalities can be considered as the condition of the Transformer decoder. The decoder aims to predict the missing latent tokens based on the condition and latent tokens: $P(z_k|z_{<k}, c)$, where z_k is the k-th tokens and $z_{<k}$ is token sequence before the k-th token. Following MASS (Song et al., 2019), we only decode the masked tokens which can be located by m' in DF-VQGAN. The training objective of MP-S2S is:

$$\mathcal{L}_2 = -\sum_k \log P(z_k|z_{<k}, c). \tag{13}$$

3.4 Inference Pipeline

As shown in Fig. 2, we use defective image x as the input image of DF-VQGAN. Since ground-truth inpainted image y cannot be accessed during inference, we use defect-free encoder by setting $m_0 = m$ and $h_0 = x$ to obtain latent tokens z^x of defective image x. Then, we use MP-S2S to predict latent tokens indicated by m' in z^x, based on the guidance of text t, defective image x, and its latent tokens z^x. Predicted tokens $\hat{z}$ will replaced the tokens in z^x by $\hat{z}' = z^x \odot (1 - m') + \hat{z} \odot m'$. The inpainted result can be reconstructed by the defect-free decoder in DF-VQGAN with $\hat{h}_T = \hat{z}'$.

4 Experiments

4.1 Implementation Details

The final pre-trained model has a total of 1.7 billion parameters. We pre-train the whole NÜWA-LIP with 64 A100 GPUs for two weeks in total. Each image is resized to 256×256. We randomly generate mask metrics m with the mask ratio of $40\% - 60\%$ in training[4]. AdamW(Kingma & Ba, 2014) optimizer is used with the warm-up ratio of 5% and dropout of 10% for both pre-training and fine-tuning. We pre-train DF-VQGAN on ImageNet(Deng et al., 2009) and MP-S2S on Conceptual Captions(Sharma et al., 2018). Please find more details of implementation in Appendix A.2.

4.2 Experiments Setup

Datasets Description In order to evaluate the proposed model, we built up three evaluation datasets, MaskCOCO, MaskFlickr, and MaskVG based on MSCOCO (Lin et al., 2014), Flickr (Young et al., 2014) and VG (Krishna et al., 2017), respectively. Different from domain-specific datasets, all of them are open-domain datasets and contain diverse language descriptions. The mask region of each image is not fixed. More details can be found in Appendix A.3.

Evaluation Metric To evaluate the quality of the inpainted image under the language guidance, the FID score (Heusel et al., 2017) is chosen as the metric. We evaluate the representation similarity of

[4]We tried different ratios in training and selected the best, see Appendix. A.6.

Table 1: **Overall results of language-guided image inpainting compared with open-domain pre-trained models.** The improvement is statistically significant with $p < 0.01$ under t-test.

MODEL	MASKCOCO				MASKFLICKR				MASKVG			
	FID$^{\downarrow}$	PSNR$^{\uparrow}$	LPIPS$^{\downarrow}$	CS$^{\uparrow}$	FID$^{\downarrow}$	PSNR$^{\uparrow}$	LPIPS$^{\downarrow}$	CS$^{\uparrow}$	FID$^{\downarrow}$	PSNR$^{\uparrow}$	LPIPS$^{\downarrow}$	CS$^{\uparrow}$
GLIDE(NICHOL ET AL., 2021)	13.5	17.25	0.233	28.99	51.9	17.06	0.225	29.16	9.0	18.53	0.210	24.94
NÜWA(WU ET AL., 2021A)	21.4	12.91	0.435	28.10	59.5	13.44	0.361	29.29	18.5	14.04	0.362	25.14
NÜWA-P	20.6	14.39	0.323	28.64	54.2	14.89	0.295	29.55	17.7	16.25	0.301	24.78
NÜWA-LIP	**12.0**	**17.35**	**0.233**	**29.39**	**42.5**	**17.47**	**0.220**	**30.76**	**8.5**	**18.70**	**0.209**	**25.05**
NÜWA-LIP (FINETUNE)	**10.5**	17.23	**0.231**	**29.65**	-	-	-	-	-	-	-	-

concepts in the inpainted and original images. We use CLIP Score (CS) to measure language-vision consistency and PSNR or LPIPS for extra quantitative experiments.

Baselines We compare NÜWA-LIP with two strong baselines. GLIDE (Nichol et al., 2021) is an effective diffusion-based model for image generation and editing. We use the public version of GLIDE in the official repository. NÜWA (Wu et al., 2021a) is another effective model for vision generation and editing. We re-implement NÜWA and re-train NÜWA following their description in the paper. Additionally, we modified NÜWA to NÜWA-P by pasting the non-defective region on the inpainting result to make a comprehensive comparison.

4.3 OVERALL RESULTS

We first compare NÜWA-LIP with the two baseline models without fine-tuning. Referring to Tab. 1, we can observe NÜWA-LIP achieves state-of-the-art performance on all datasets. Our proposed method outperforms the highest baseline GLIDE with 1.5 FID on MaskCOCO, 9.4 FID in MaskFlickr, and 0.5 on MaskVG, which shows the effectiveness of NÜWA-LIP. We suggest that the improvement is ascribed to that NÜWA-LIP can not only remain non-defective region unchanged but also avoid inaccurate or incomprehensive encoding. We conduct human evaluation in Appendix A.4.

4.4 COMPARISON WITH OTHER MODELS

To reveal the effectiveness of NÜWA-LIP, we conduct extra experiments with other inpainting models. TDANET (Zhang et al., 2020) and AMLR (Wu et al., 2021b) are two non-pre-trained language-guided inpainting models. MASKGIT (Chang et al., 2022) and LAMA (Suvorov et al., 2022) are class-conditional and unconditional inpainting model pre-trained on large-scale data. For MASKGIT, we use CLIP to classify the class of the ground-truth image as the input. In Tab. 2, NÜWA-LIP outperforms all these models, which shows its effectiveness and the essentials of the language.

STABLE DIFFUSION (Rombach et al., 2022) is an effective model on visual synthesis tasks. However, the setting of STABLE DIFFUSION is different from most prior image inpainting works (Zhang et al., 2020; Suvorov et al., 2022; Chang et al., 2022). In general image inpainting setting, the input image for the inpainting model is defective or damaged. However, the input image of STABLE DIFFUSION needs to be a normal image without defective or damaged regions, which is called image editing in most related works. For STABLE DIFFUSION (IMAGE EDITING) in Tab. 2, the input image will be the ground-truth image. For STABLE DIFFUSION in Tab. 2, which means image inpainting setting like ours and other baselines, the input image will be the defective image. Moreover, STABLE DIFFUSION trained on LAION-5B dataset, which is $1000\times$ larger than ours. Even though, we not only outperform STABLE DIFFUSION's image inpainting results but also obtain comparable performance with its image editing results. More explanations are in Appendix A.5.

4.5 EFFECTIVENESS OF DF-VQGAN

To reveal the advantages of DF-VQGAN, we train DF-VQGAN and VQGAN on the same training steps and data on two different tasks. We use the official implementation of VQGAN for fairness.

Image Reconstruction The goal of this task is to transform each complete image into discrete latent tokens and then reconstruct the image based on these discrete tokens. This task can assess the image encoding and decoding performance of VQGAN, which are critical to NÜWA-LIP. Surprisingly, from Tab. 3, one can see that DF-VQGAN performs better than VQGAN in the same setting (*i.e.*, resolution and vocab size). This improvement may be ascribed to that the training of DF-VQGAN

Table 2: **Comparsion with different models on MaskCOCO.** We achieved state-of-the-art performance. † denotes trained or finetuned on COCO.

MODEL	FID↓	CLIP SCORE↑
STABLE DIFFUSION (IMAGE EDITING)	10.9	30.38
STABLE DIFFUSION	54.2	27.19
TDANET†	27.2	27.90
AMLR†	24.7	27.41
LAMA	17.3	24.38
MASKGIT	15.5	27.20
NÜWA-LIP (W/O PRETRAIN)†	**11.0**	**28.74**
NÜWA-LIP	**12.0**	**29.34**
NÜWA-LIP (FINETUNE)†	**10.5**	**29.65**

Table 3: **Comparsion of DF-VQGAN and VQGAN.** DF-VQGAN beats VQGAN on both image reconstruction (IMG.REC), oracle inpainting (ORC.INP), and average distance (AVG.DIS) under the same setting.

MODEL	RESOLUTION	LENGTH	VOCAB	IMG.REC	ORC.INP	AVG.DIS
VQGAN	$256^2 \rightarrow 16^2$	256	1024	12.47	16.30	5.75
DF-VQGAN	$256^2 \rightarrow 16^2$	256	1024	**11.16**	**5.56**	**4.54**
VQGAN	$256^2 \rightarrow 16^2$	256	12288	5.48	7.15	9.77
DF-VQGAN	$256^2 \rightarrow 16^2$	256	12288	**5.16**	**2.95**	**4.31**
VQGAN	$256^2 \rightarrow 32^2$	1024	8192	1.47	2.04	16.93
DF-VQGAN	$256^2 \rightarrow 32^2$	1024	8192	**1.38**	**0.80**	**3.52**

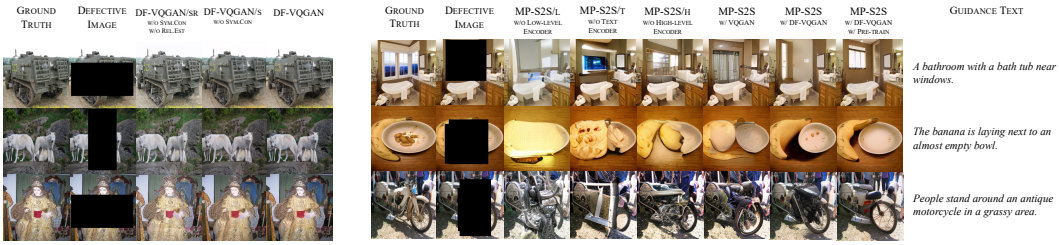


Figure 5: **Qualitative ablations of proposed components for DF-VQGAN and MP-S2S.** We can observe that the model achieves the best performance combing with all proposed components.

covers both complete image reconstruction and incomplete image reconstruction illustrated in Fig. 3, which makes the resulting model more robust.

Oracle Inpainting This task aims to reconstruct the image given the ground-truth discrete tokens of the defective region (encoded from the ground-truth image) and the discrete tokens of the rest region (encoded from the defective image). From Tab. 3, one can see that DF-VQGAN performs better than VQGAN in the same setting (*i.e.*, resolution and vocab size), which verifies the effectiveness of DF-VQGAN for the image inpainting task. Fig. 4 shows some visualization examples of VQGAN and DF-VQGAN on this task.

Average Distance To verify whether DF-VQGAN avoids fusing uncertain information from defective regions, we measure the consistency of z^x and z^y. For each defective x and non-defective image y pair, we obtain their representations z^x and z^y. We only measure the known part which is indicated by mask m. Then we calculate the average vector distance as the measure of their consistency. We find DF-VQGAN improves the consistency of z^x and z^y largely. This can be interpreted as the mathematical reason for the effectiveness of DF-VQGAN.

In Fig. 4, we compare DF-VQGAN and VQGAN with VQGAN-P, which copies the non-defective region from the input image and uses it to substitute the reconstruction part of the same region by VQGAN. In comparison to VQGAN-P, DF-VQGAN has a significantly better transition of the non-defective region and inpainted region. More quantitative comparisons are in Appendix. A.7.

4.6 ABLATION STUDIES

DF-VQGAN The ablation of DF-VQGAN aims to verify whether the ***symmetrical connection*** and ***relative estimation*** play the expected roles. DF-VQGAN/SR and DF-VQGAN/S respectively denote the model without both symmetrical connection and relative estimation or symmetrical connection. We re-train DF-VQGAN under different settings and evaluate oracle inpainting mentioned in Sec 4.5 on ImageNet. Referring to Tab. 4, we can observe that ***relative estimation*** reduces the FID by 2.31 and the ***symmetrical connection*** further provides a decrease of 2.49.

MP-S2S To verify whether the multi-perspective benefits language-guided image inpainting, we conduct the ablation on MP-S2S by eliminating one or two of its perspectives. MP-S2S/L, MP-S2S/H, or MP-S2S/T denotes the MP-S2S model without low-level pixel encoder, high-level token encoder or text encoder. MP-S2S/HL denotes the MP-S2S model which only has a text encoder.

Table 4: **Ablation of DF-VQGAN on ImageNet.** We can observe that both ***relative estimation*** (REL.EST) and ***symmetrical connection*** (SYM.CON) improves the performance of DF-VQGAN significantly.

MODEL	COMPONENT		FID↓
	SYM.CON	REL.EST	
DF-VQGAN/SR	×	×	7.15
DF-VQGAN/S	×	√	5.44
DF-VQGAN	√	√	**2.95**

Table 5: **Ablation of MP-S2S on MaskCOCO.** Text encoder (TEXT), high-level token encoder (HIGH) and low-level pixel encoder (LOW) understand the language and defective image from multi perspectives to improve the performance.

MODEL	COMPONENT			FID↓
	TEXT	HIGH	LOW	
MP-S2S/HL	√	×	×	29.4
MP-S2S/L	√	√	×	27.4
MP-S2S/H	√	×	√	26.8
MP-S2S/T	×	√	√	34.7
MP-S2S	√	√	√	**26.2**
+DF-VQGAN	√	√	√	**11.0**

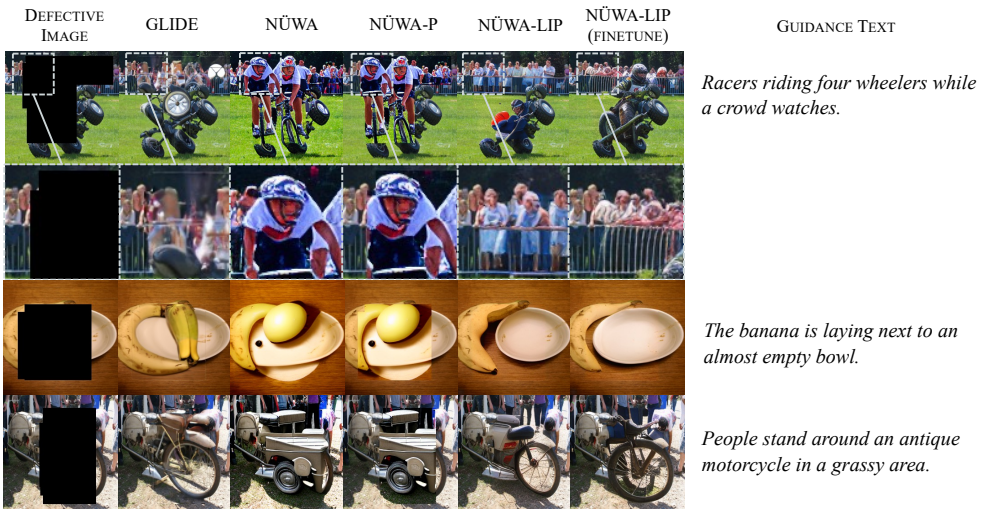


Figure 6: **Language-guided image inpainting results of different models.** NÜWA-LIP achieves the best quality compared with baselines. More cases are in Supplementary Materials.

To exclude the influence from other components, we use the same original VQGAN as a backbone. We re-train the MP-S2S under different settings on MSCOCO and evaluate the performance of language-guided image inpainting results on MaskCOCO. Referring to Tab. 5, one can observe the text perspective provide the highest gain of 8.5, and high-level tokens and low-level pixels perspective gains the model with FID of 1.2 and 0.6, respectively. When we combine DF-VQGAN with MP-S2S, we obtain a huge improvement of 15.2 on FID, which shows the capability of the whole framework.

4.7 CASE STUDIES

We selected several cases to show the inpainting results. Referring to Fig. 6, one can observe that (1) compared with NÜWA, NÜWA-LIP remains the correct hue of the whole image, (2) compared with baselines, NÜWA-LIP makes a better transition of non-defective and inpainted region, (3) NÜWA-LIP generates more details which accord with the guidance text, and (4) fine-tuning NÜWA-LIP further improves the quality of the inpainting. We also provide samples of free-form mask inpainting and out-of-domain image inpainting. See Appendix A.9 for more details.

5 CONCLUSION

In this paper, we proposed NÜWA-LIP for language-guided image inpainting. Our contributions are three-fold: (1) We proposed DF-VQGAN, which can control receptive spreading and protect information. (2) We proposed MP-S2S, which further enhances visual information from complementary perspectives. (3) We built 3 open-domain benchmarks, where NÜWA-LIP is also superior to baselines. We analyze the broader impact and limitations of NÜWA-LIP in Appendix A.11.

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A APPENDIX

A.1 DETAILED DESCRIPTION OF DEFECT-FREE ENCODER

In Defect-free Encoder, all normal operations will be replaced with defect-free operations, which are not only normalization but also convolution and attention. In defect-free Encoder, for convolution, we masked the defective region to zero padding. For attention, we force the attentive score of defective regions to zero to avoid getting information from there.

A.2 IMPLEMENTATION DETAILS OF DF-VQGAN AND MP-S2S

DF-VQGAN DF-VQGAN uses the same architectures and hyperparameters as VQGAN. We set the vocab size to $8,192$, the learning rate to 5×10^{-6}, and the batch size to 200. The dim of the latent token is 256. The resolution is $256^2 \rightarrow 32^2$, where the discrete token is corresponding to the patch size of 8×8. We use ImageNet (Deng et al., 2009) as the pre-training corpus.

MP-S2S We set the number of layers in encoder E^{l}, E^{h} and E^{t} to 12, and the number of layers in the autoregressive decoder to 24. All Transformers have 20 heads and the hidden size is 1024. We set the learning rate to 5×10^{-4}, and the batch size to 320. The text encoder E^{t} and tokenizer is initialized with the text encoder and tokenizer of CLIP (Radford et al., 2021). The E^{l} and E^{h} are trained from scratch. We use Conceptual Captions (Sharma et al., 2018) as the pre-training corpus.

A.3 DATASET DETAILS

We follow Karpathy split (Karpathy & Fei-Fei, 2015) and select the test split of MSCOCO and Flickr to build MaskCOCO and MaskFlickr. For MaskVG, we sample $10,000$ samples randomly from the VG dataset. For each image-text pair, the original image and corresponding caption are considered as full image and guidance text. Each image will be cropped and resized to the resolution of 256×256. The defective image is generated by masking either one bounding box of the object or a random irregular region in the original image. The statistics of the datasets are listed in Tab. 6. We will release them under a Creative Commons Attribution 4.0 License.

Table 6: **Statistics of the evaluation datasets.** The defective image is generated by masking either one bounding box of the object or a random irregular region.

DATASET	IMAGE-TEXT PAIRS	MASK RATIO
MASKCOCO	5000	31.5%
MASKFLICKR	1000	48.3%
MASKVG	10000	14.6%

A.4 HUMAN EVALUATION

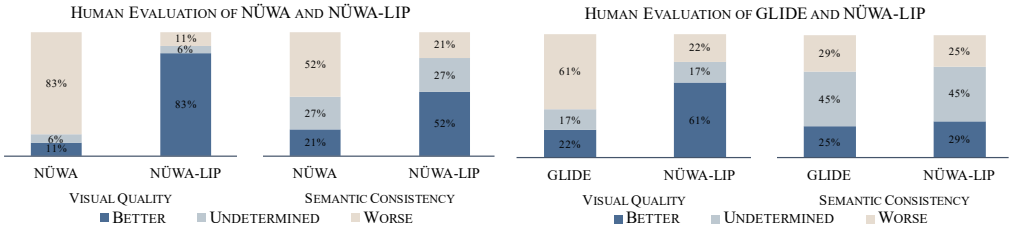


Figure 7: **Human evaluation of NÜWA-LIP and baselines.** NÜWA-LIP obtains better visual quality and semantic consistency in human evaluation.

To further compare the quality of NÜWA-LIP and baselines, we evaluate the inpainting results by humans. We randomly sample 500 samples from the MaskCOCO dataset and ask human annotators to compare the visual quality and semantic consistency of NÜWA-LIP, NÜWA, and GLIDE with

random order and hiding the model's name. The visual quality represents the quality of the entirety, including the transition of the edge or continuity of pixels. The semantic consistency represents the compliance of the language guidance. Referring to Fig. 7, NÜWA-LIP's visual quality increased significantly in comparison with NÜWA or GLIDE. We also achieve better semantic consistency.

A.5 Comparison with Stable Diffusion

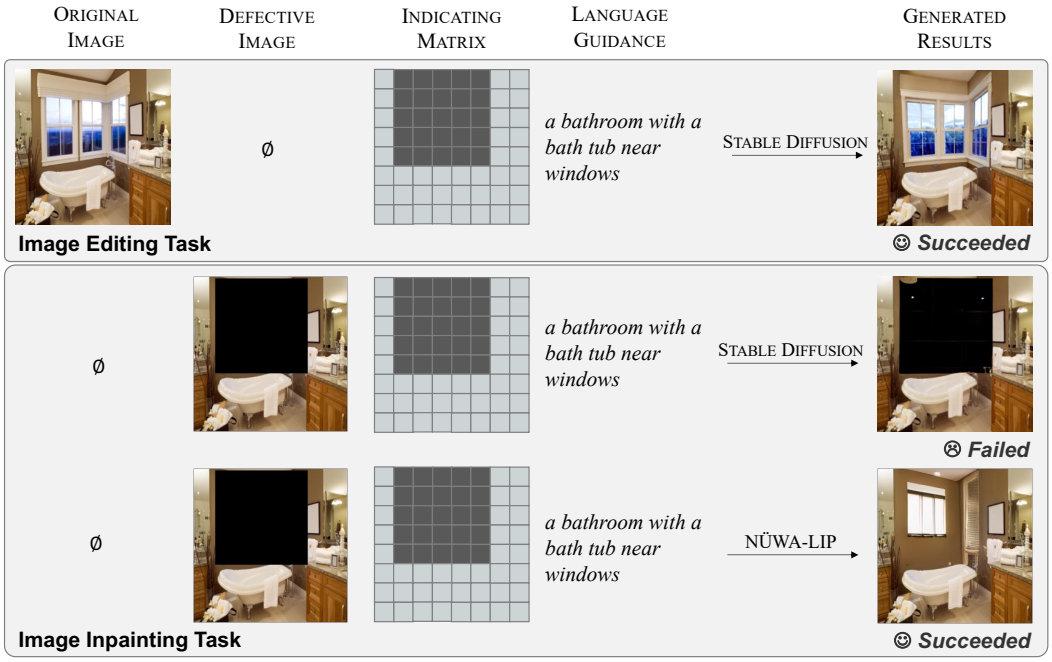


Figure 8: **Comparison of Stable Diffusion and NÜWA-LIP pipelines.** Different from most prior works, the input image of Stable Diffusion needs to be well-formed, which is called image editing in most related works.

Stable Diffusion is an effective model for visual synthesis tasks, which shows attractive performance in the generation. As Fig. 8, the setting of Stable Diffusion is different from most prior image inpainting works (Zhang et al., 2020; Suvorov et al., 2022; Chang et al., 2022). In general image inpainting setting, the input image for the inpainting model is defective or damaged. However, the input image of Stable Diffusion needs to use normal images without defective or damaged regions, which is called image editing in most related works (Nichol et al., 2021; Wu et al., 2021a; Chang et al., 2022). For Stable Diffusion (Image Editing) in Tab. 2 and the first row of Fig. 8, the input image will be the ground-truth image. For Stable Diffusion in Tab. 2 and the second row of Fig. 8, which means image inpainting setting like ours and other baselines, the input image will be the defective image. We find that Stable Diffusion is hard to handle images with defective or damaged regions. As Fig. 8, the inpainting of Stable Diffusion may fail in inpainting setting. We use their official implementation from Diffusers[5]and checkpoints[6]. We do not change any hyper-parameter during inference. We explained this as the receptive spreading of defective regions and the training-inference gap of the model.

A.6 Mask Ratio of DF-VQGAN

We conduct an ablation study of different mask ratios during training DF-VQGAN. Tab. 9 shows FID scores with different mask ratios evaluated on ImageNet. In this work, we use the best $40\% - 60\%$ ratio with an FID of 2.95 during training DF-VQGAN.

[5]https://github.com/huggingface/diffusers

[6]We use the best sd-v1-4 checkpoint.

Table 7: **Ablation of mask ratio in oracle inpainting on ImageNet.** We select the best mask ratio to train our DF-VQGAN.

MASK RATIO	RESOLUTION	TOKEN LENGTH	VOCAB SIZE	FID↓
$20\% - 40\%$	$256^2 \rightarrow 16^2$	256	12288	3.07
$40\% - 60\%$	$256^2 \rightarrow 16^2$	256	12288	**2.95**
$60\% - 80\%$	$256^2 \rightarrow 16^2$	256	12288	3.18

A.7 MORE QUANTITATIVE COMPARISONS OF DF-VQGAN

Table 8: **More quantitative comparisons of DF-VQGAN on ImageNet.** DF-VQGAN beats VQGAN or VQGAN-P on both image reconstruction (IMG.REC) and oracle inpainting (ORC.INP).

MODEL	RESOLUTION	TOKEN LENGTH	VOCAB SIZE	IMG.REC	ORC.INP
VQGAN	$256^2 \rightarrow 16^2$	256	12288	6.03	7.15
VQGAN-P	$256^2 \rightarrow 16^2$	256	12288	6.03	3.77
TS-VQGAN	$256^2 \rightarrow 16^2$	256	12288	5.89	6.47
DF-VQGAN/S	$256^2 \rightarrow 16^2$	256	12288	**5.14**	**5.44**
DF-VQGAN	$256^2 \rightarrow 16^2$	256	12288	**5.16**	**2.95**

To figure out whether ***relative estimation*** can avoid receptive spreading of defective regions, we compare VQGAN with DF-VQGAN/S, which is DF-VQGAN without ***symmentrical connection***. In the upper part of Tab. 8, we can find that we significantly increase the FID of 5.44 v.s. 7.15 in oracle inpainting task. Meanwhile, the gain in image reconstruction can be observed which can be explained by ***relative estimation*** improving the robustness of the model.

To figure out whether ***symmentrical connection*** can protect the information of non-defective regions, we compare DF-VQGAN with VQGAN-P, which directly copy and paste the non-defective region of the image. In the bottom part of Tab. 8, we can find that we still obtained a better FID of 2.95 v.s. 3.77 in oracle inpainting task. This can be explained that we can make a significantly better transition of the non-defective region and inpainted region (see Fig. 4). Meanwhile, we still keep the comparable FID of 5.16 in the image reconstruction task compared DF-VQGAN/S. This shows the good cooperation of ***relative estimation*** and ***symmentrical connection*** we proposed.

We also conduct the comparison with TS-VQGAN (Cao et al., 2021), which is used in conditional image synthesis to encode an image without defective regions. The goal of TS-VQGAN is to avoid information leaking, which means results are more like reference images instead of the target condition. Different from TS-VQGAN, DF-VQGAN works in the image inpainting scenario, which means defective regions and non-defective regions exist at the same time in an image. Thus, we can observe that we outperform TS-VQGAN with significant improvement in oracle inpainting.

A.8 DIFFERENT ENCODER AND DECODER LAYERS OF MP-S2S

Table 9: **Ablation of mask ratio in oracle inpainting on ImageNet.** We select the best mask ratio to train our DF-VQGAN.

MODEL	ENCODE LAYER	DECODE LAYER	FID↓
MP-S2S/A	24	12	26.7
MP-S2S/B	12	12	28.1
MP-S2S	12	24	**26.2**

We conduct a comparison of different encoder and decoder layers. We can observe that decoder layers are more important than encoder layers. The decoder of MP-S2S needs to decode based on low-level pixels, high-level tokens, and language encoders. Some works, like Parti (Yu et al., 2022) also reported that decoder layers are more important than encoder layers for autoregressive models.

A.9 FREE-FORM MASK INPAINTING AND OUT-OF-DOMAIN IMAGE INPAINTING

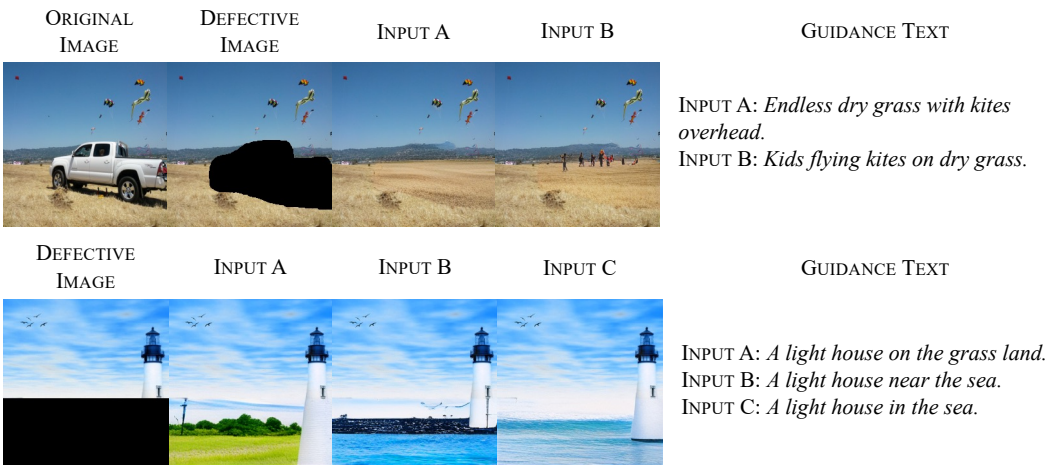


Figure 9: **Free-form mask inpainting and out-of-domain image inpainting.** The results show NÜWA-LIP's good generalization ability and effectiveness on open-domain data.

In Fig. 9, NÜWA-LIP does not train on the free-from mask, it performs well on free-form mask inpainting. We also show the result of the out-of-domain image. The results show NÜWA-LIP's good generalization ability and effectiveness on open-domain data.

A.10 FAILURE CASE

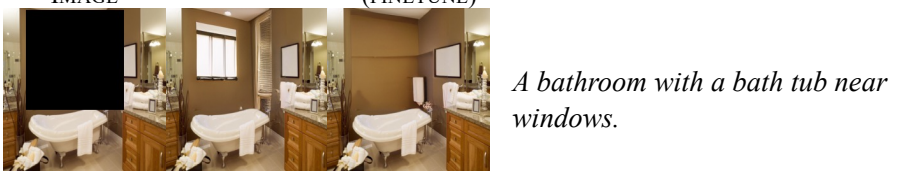


Figure 10: **Failure case of NÜWA-LIP.** We explained this as fine-tuning leads to the forgetting of rare objects in the fine-tuning dataset.

Although NÜWA-LIP shows effectiveness in most cases, we still find in some cases like Fig. 10, that show fine-tuning will cause failure in inpainting some objects. This occurred in a very low frequency and in most cases, fine-tuning will increase the quality of the inpainted image significantly. We explained this as fine-tuning leads to the forgetting of rare objects in the fine-tuning dataset.

A.11 BROADER IMPACT AND LIMITATIONS

NÜWA-LIP, which is an effective model for language-guided image inpainting, can provide the potential for creators and motivate their creativity. For example, it will be convenient for creators to modify or fix the image under the guidance of their natural language. However, fake images may be abused for some scenes, such as news reporting. As the common issue of autoregressive models, handling an extremely large image may lead to much higher computational cost.